Khawaja Abaid Ullah

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PhD researcher at RIT's KAS Lab working on multi-agent coordination and decision-making for autonomous systems with goal of building autonomous robot teams for disaster search and rescue. Additionally research machine learning security. Open-source contributor to Keras 3. Creator of Teras, an open-source tabular DL library that supports JAX, TensorFlow and PyTorch backends. Former AWS AI/ML intern.

Education

PhD in Electrical and Computer Engineering **Jun 2026 - Expected May 2029**
Rochester Institute of Technology *Rochester, NY*
• Fully funded Graduate Research Assistantship. Advised by Dr. Mohammad Javad Khojasteh, Khojasteh Autonomous Systems Laboratory.

M.S. in Artificial Intelligence **Aug 2024 - May 2026**
Rochester Institute of Technology *Rochester, NY*
• GPA: 4.0/4.0. Fully funded by the U.S. Department of State Fulbright Scholarship.
• Thesis: "Antidistillation Sampling for Classification."

B.S. in Computer Science **2017 - 2021**
University of Narowal *Narowal, Pakistan*
• Graduated first in the Computer Science class of 2021.

Selected Papers & Preprints

K. A. Ullah, M. J. Khojasteh. *"ADS-C: Antidistillation Sampling for Classification."* arXiv:2607.15467, 2026. Under review, IEEE Open Journal of the Computer Society. Paper | Code
• Proposed a margin-aware inference-time defense that preserves every served top-1 prediction while reducing distilled-student accuracy by 29.6 pp on CIFAR-10, 17.4 pp on CIFAR-100, and 13.3 pp on Tiny-ImageNet.
• Released the defense and its evaluation harness as open-source PyTorch code, enabling reproduction across all three benchmarks and multiple student architectures.

K. A. Ullah, M. J. Khojasteh. *"Reversing the Incentive to Distill: A Review of Test-Time Antidistillation Defenses for Classification."* Under review, 2026.
• Built a unified matched-cost benchmark and a method-agnostic top-1-preservation wrapper, directed perturbations drove distilled students 34.5 pp (CIFAR-10) and 47.7 pp (CIFAR-100) below the hard-label baseline.
• Reimplemented prior test-time defenses under a common threat model and matched-cost evaluation protocol to isolate which perturbation mechanisms genuinely suppress student learning.

H. Mahmud, **K. A. Ullah**, M. J. Khojasteh, J. Heard, P. David. *"How Do Large Language Models Judge Social Attraction? Evidence from Theory-Grounded Persona Ratings Across Multiple LLMs and Humans."* arXiv:2608.09717, 2026. Under review.
Paper | Code
• Evaluated social-judgment reliability and human-LLM alignment across 34 LLMs and 198 human participants, including controlled analyses of gender presentation.

Research Experience

Graduate Research Assistant - Khojasteh Autonomous Systems Laboratory **Jun 2026 - Present**
Rochester Institute of Technology *Rochester, NY*
• Research multi-agent coordination and decision-making for autonomous systems, with current projects in dynamic resource allocation, effective multiagent communication under various constraints, as well as detection and containment of compromised agents, with the final goal towards autonomous robot teams for disaster search and rescue.
• Research machine learning security specifically antidistillation approaches to prevent unauthorized knowledge distillation of deployed machine learning models.

Industry Experience

Software Development Engineer, AI/ML Intern **May 2025 - Aug 2025**
Amazon Web Services (AWS) *East Palo Alto, CA*
• Engineered core Apache Spark testing modules for a next-generation internal benchmarking platform, closing feature-parity gaps with its legacy predecessor across targeted benchmarking workflows.
• Exceeded the original project scope by independently identifying and implementing additional benchmarking capabilities, earning strong recognition from team leadership and a full-time return offer.

Junior Python Developer (Remote) **Feb 2023 - May 2023**
Comundo *Copenhagen, Denmark*
• Built semi-supervised web-scraping and data-curation pipelines for energy-consumption data; cleaned and aggregated outputs for CO2 estimation and downstream ML forecasting.

Open-Source Contributions

Open-Source Developer **May 2022 - Aug 2024**
Independent

Teras - GitHub

• Created a Keras 3-based tabular deep-learning library supporting JAX, PyTorch, and TensorFlow. Implemented state of the art research architectures for classification, regression, data imputation, and synthetic-data generation.

Keras

• Contributed Beta and Binomial random-sampling functionality to the multi-backend Keras 3 API across JAX, PyTorch, and TensorFlow.

micrograd_c - GitHub

• Implemented Andrej Karpathy's micrograd from scratch in C: a scalar autograd engine and neural-network library with a PyTorch-like API.

Selected Research Projects

Fontus - Multi-Agent RL for Aerial UAV Recharging **Dec 2025**

• Built a custom PettingZoo Parallel multi-agent environment with vectorized NumPy dynamics, energy-aware transitions, and proximity-triggered charging and trained parameter-shared IPPO/PPO policies with PyTorch and Ray RLlib.

• Established an analytical no-charge baseline and diagnosed a reward-hacking "lazy agent" local optimum: apparent reward gains reflected passive survival rather than successful refueling, motivating off-policy/replay-based exploration.

codriver - Resource-Efficient Contextual Code Generation **Dec 2024**

• Inspired by GitHub copilot, built a custom 260M-parameter contextual code-generation tool based on Google Deepmind's Gemma LLMs.

• Ported Gemma2 from Flax linen to NNX API, and applied staged knowledge distillation first using Gemma 2 2b and then CodeGemma 2b as teacher models, using text and code datasets respectively for the model to learn lexical as well as code structure.

Technical Skills

Languages: Python, C, C++, Bash
ML / AI: JAX, Flax, Keras, PyTorch, TensorFlow, NumPy
Systems / Tooling: Linux, Docker, Git, ROS 1, ROS 2, Arduino
Methods: Reinforcement Learning, Multi-Agent RL, Multi-Agent Systems, Neural Combinatorial Optimization, LLM-Assisted Optimization, Knowledge Distillation, Model Extraction

Teaching, Service & Honors

Teaching Assistant - Principles of Robotics **Aug 2026 - Present**
Rochester Institute of Technology *Rochester, NY*

• Develop lecture and laboratory material for a combined undergraduate and graduate robotics course; labs use ROS 1, ROS 2, and Arduino.

• Designed and built the course's ROS based docker containers and reproducible VM images, replacing a brittle third-party dependency for the full lab sequence.

Reviewer, IEEE Signal Processing Letters **Jun 2026 - Present**

• Review manuscripts in machine learning, signal processing, and related areas for technical soundness, novelty, experimental methodology, and reproducibility.

Honors: Fulbright Scholar (U.S. Department of State); Kaggle Expert.